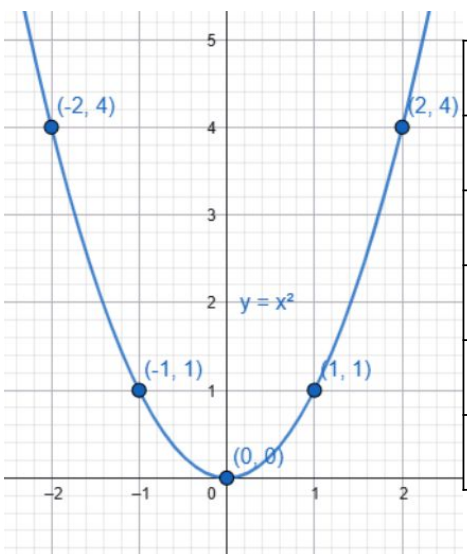
APPLIED MATHEMATICS LABORATORY

3D SURFACES

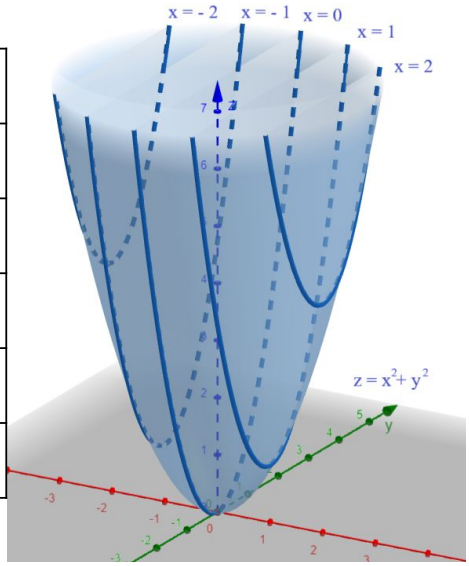
SURFACE TRACES

When we have an equation with two variables, we can graph it on a Cartesian plane, where the first variable is graphed on the x-axis and the second variable is graphed on the y-axis. This gives us a curve. When we have an equation with three variables, we can graph it on a Cartesian plane that has three axes. This gives us a surface.

A helpful way to understand the shape of a curve or surface is to choose values for one of the variables and look at the equation for those values. The result is an equation with one less dimension, a one-dimensional point or a two-dimensional trace. By looking at several points or traces, we can gain a good sense of what the shape of the curve or surface is.



Point	x	Trace
$y = 4$	2	$z = y^2 + 4$
$y = 1$	1	$z = y^2 + 1$
$y = 0$	0	$z = y^2$
$y = 1$	-1	$z = y^2 + 1$
$y = 4$	-2	$z = y^2 + 4$



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